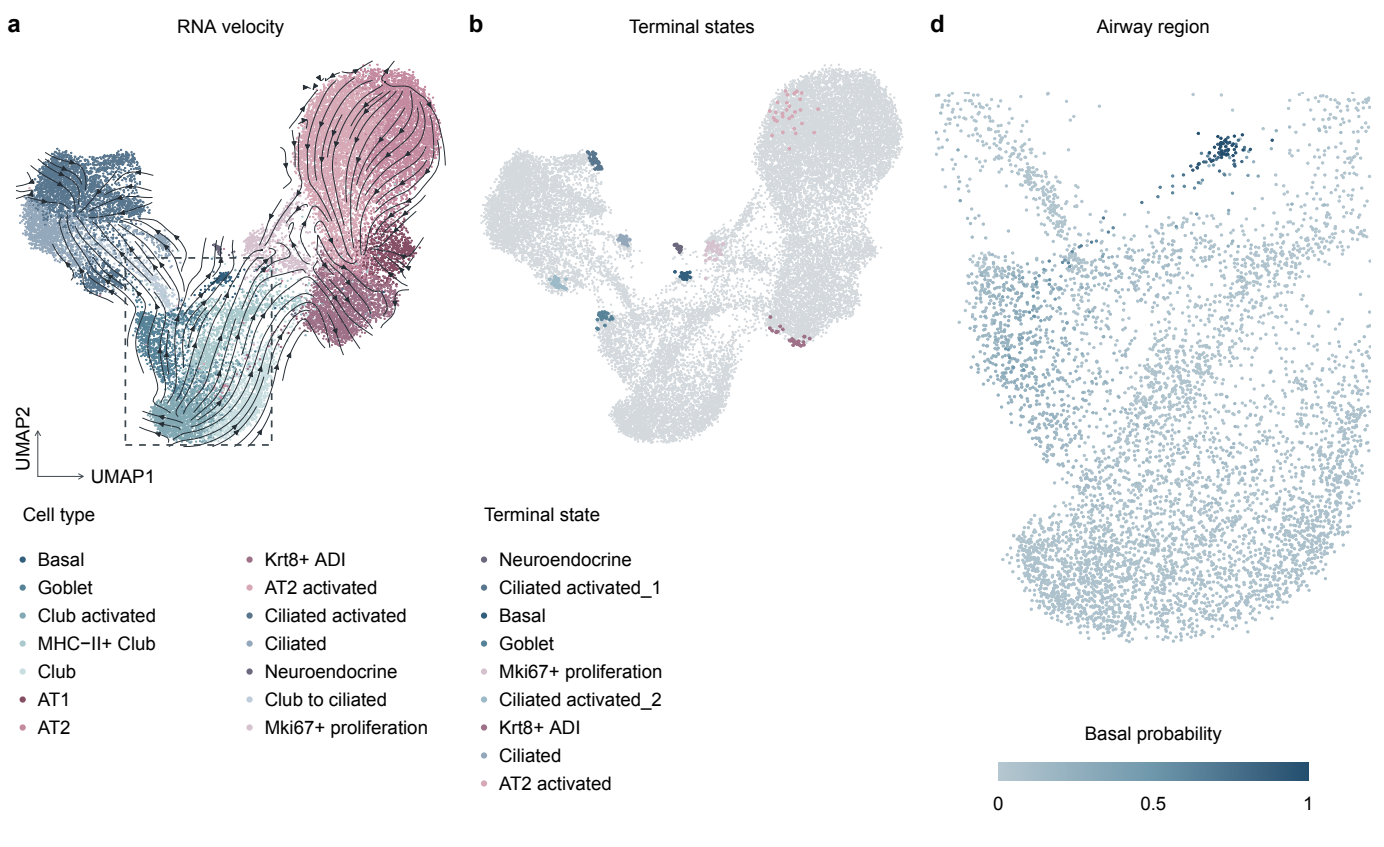

a
RNA velocity
UMAP2
UMAP1
b
Terminal states
d
Airway region
Cell type
Basal
Goblet
Club activated
MHC−II+ Club
Club
AT1
AT2
Krt8+ ADI
AT2 activated
Ciliated activated
Ciliated
Neuroendocrine
Club to ciliated
Mki67+ proliferation
Terminal state
Neuroendocrine
Ciliated activated_1
Basal
Goblet
Mki67+ proliferation
Ciliated activated_2
Krt8+ ADI
Ciliated
AT2 activated
Basal probability
0
0.5
1